

AI Powered Legal Advisor and Analyzer



Final Year Project Proposal

Design Project (Part-I)

Team Members

Muhammad Nabeel (BCS233115)

Ashir Qureshi (BCS233084)

Muhammad Sharjeel (BCS233113)

Supervised By

Ms. Khadija Iftikhar

Fall 2026

Bachelor of Science in Computer Science

Department of Computer Science

Capital University of Science & Technology, Islamabad

Approval Certificate

This project, entitled as “AI Powered Legal Advisor and Analyzer” has been approved for the award
of

Bachelors of Science in Computer Science.

Committee Signatures:

Supervisor:

(Supervisor Name Ms. Khadija Iftikhar)

Project Coordinator:

(Name)

Head of Department:

(Name)

Declaration

I/We, hereby, declare that “No portion of the work referred to, in this project has been submitted in support of an application for another degree or qualification of this or any other university/institute or other institution of learning”. It is further declared that this undergraduate project, neither as a whole nor as a part thereof has been copied out from any sources, wherever references have been provided.

MEMBER NAME	MEMBERS' SIGNATURES
Muhammad Nabeel (BCS233115)	
Ashir Qureshi (BCS233084)	
Muhammad Sharjeel (BCS233113)	

Acknowledgements

It is usual to thank those individuals who have provided particularly useful assistance, technical or otherwise, during your project. Your supervisor will obviously be pleased to be acknowledged as he or she will have invested quite a lot of time overseeing your progress.

Table of Contents

Chapter 1 — Introduction.....	7
1.1 Project Introduction.....	7
1.1.1 Project Goal.....	7
1.1.2 Problem Statement.....	7
1.1.3 Objectives.....	7
1.1.4 Proposed Solution.....	7
1.1.5 Users.....	8
1.1.6 System Workflow.....	8
1.1.7 Scope.....	9
1.1.8 Expected Benefits.....	9
1.2 Existing Examples / Solutions.....	10
1.3 Business Scope.....	11
1.4 Useful Tools and Technologies.....	12
1.5 Project Work Breakdown.....	14
1.6 Project Timeline.....	16

List of Figures

Figure 1: Main workflow of the proposed platform.....	8
Figure 2: Initial legal scope — Property Disputes and Family Disputes.....	9
Figure 3: Lawyer discovery and case progress.....	9
Figure 4: System architecture.....	13
Figure 5: RAG pipeline — offline knowledge base preparation and online answering.....	13

List of Tables

Table 1: Users of the platform and their roles.....	8
Table 2: Initial legal scope and topics covered.....	9
Table 3: Profile of MyCounsel.....	10
Table 4: Profile of QanoonAI.....	10
Table 5: Comparison of existing solutions and the proposed system. “Not stated” means the public description does not mention the feature.....	11
Table 6: Business scope of the platform.....	11
Table 7: Technology stack.....	12
Table 8: Work breakdown structure.....	14
Table 9: Team assignment.....	15
Table 10: Seven-month project timeline (shaded cells show active work).....	16

Chapter 1 — Introduction

1.1 Project Introduction

1.1.1 Project Goal

To build a case-centered AI platform that helps Pakistani citizens understand property and family legal issues using verified Pakistani legal sources, and helps them reach suitable lawyers.

1.1.2 Problem Statement

Many citizens find legal matters hard to understand. Laws, rules and court procedures are spread across different documents and are written in technical language. A person with a tenancy, inheritance, divorce or custody problem often does not know which law applies, what documents to collect, or which lawyer to contact. General-purpose AI chatbots can answer quickly, but they may not rely on verified Pakistani sources and can produce statements that cannot be checked, which is risky in a legal setting.

1.1.3 Objectives

The project has the following objectives:

1. To build a verified knowledge base of Pakistani legal sources for property and family disputes.
2. To develop a RAG-based assistant that answers from retrieved legal text and always shows its sources.
3. To ask only the follow-up questions needed to clarify a case before giving guidance.
4. To process uploaded documents (PDF and scanned images) and generate a case-specific evidence checklist.
5. To provide a verified lawyer directory with search filters, and to track each case until resolution.
6. To evaluate the system against a standard LLM baseline using defined quality and usability measures.

1.1.4 Proposed Solution

The platform combines three parts:

- **A RAG-based legal assistant.** It answers from a verified legal knowledge base rather than from the model's memory. Every answer is built from retrieved legal text, and the sources are shown to the user so that the answer can be checked.
- **A verified lawyer directory.** Lawyer profiles are verified by an administrator, and users can narrow their search by filters such as specialization and city.
- **Case tracking.** Each case is followed from the first query to its resolution, so the user always knows the current status and the next step.

1.1.5 Users

Table 1: Users of the platform and their roles

User	Role on the platform
Citizen	Main user. Creates cases, answers follow-up questions, uploads documents, reads guidance, searches lawyers and follows case progress.
Lawyer	Maintains a professional profile and accesses cases that a citizen has chosen to share with them.
Administrator	Manages users, lawyers, cases and legal content, and verifies lawyers before they appear in the directory.

1.1.6 System Workflow

The workflow is shown in Figure 1 and has nine steps, each described below.

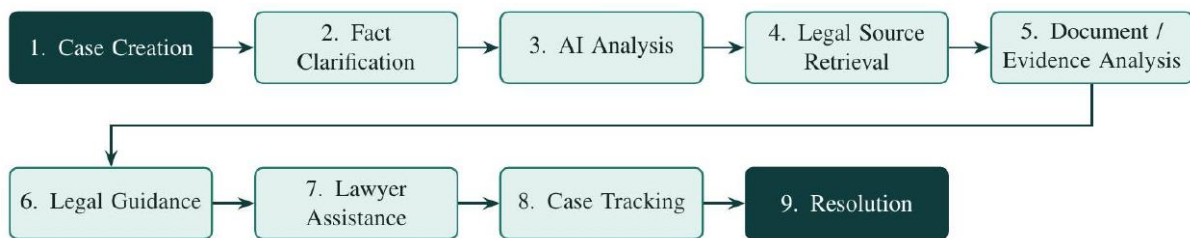


Figure 1: Main workflow of the proposed platform

1. **Case creation.** The user describes the problem in simple words.
2. **Fact clarification.** The system asks only the follow-up questions that are needed, for example whether a tenancy agreement exists or whether a marriage is registered.
3. **AI analysis.** The system analyzes the collected facts to decide which legal topics are involved.
4. **Legal source retrieval.** The RAG component retrieves the most relevant legal sources from the knowledge base (Figure 5).
5. **Document / evidence analysis.** Uploaded documents such as agreements or notices are processed. PyMuPDF extracts text from PDFs, OCR is used for scanned files and images, and the system then extracts the relevant text and clauses.
6. **Legal guidance.** The system generates an understandable explanation with citations and a checklist of documents and evidence to prepare.
7. **Lawyer assistance.** The user searches for a suitable lawyer (Figure 3) and shares the case with consent.
8. **Case tracking.** The user follows the status of the case on the platform.
9. **Resolution.** The case is closed once it is resolved.

As noted under Figure 1, the citizen performs steps 1, 2, 5, 6 and 8; the AI/RAG component performs steps 3 to 5; the lawyer is involved in step 7; and the administrator monitors the whole workflow.

1.1.7 Scope

The initial system covers only the two areas shown in Figure 2. No other legal area is included in this project. Limiting the scope allows the team to build a smaller, higher-quality and verifiable knowledge base instead of a broad but unreliable one.

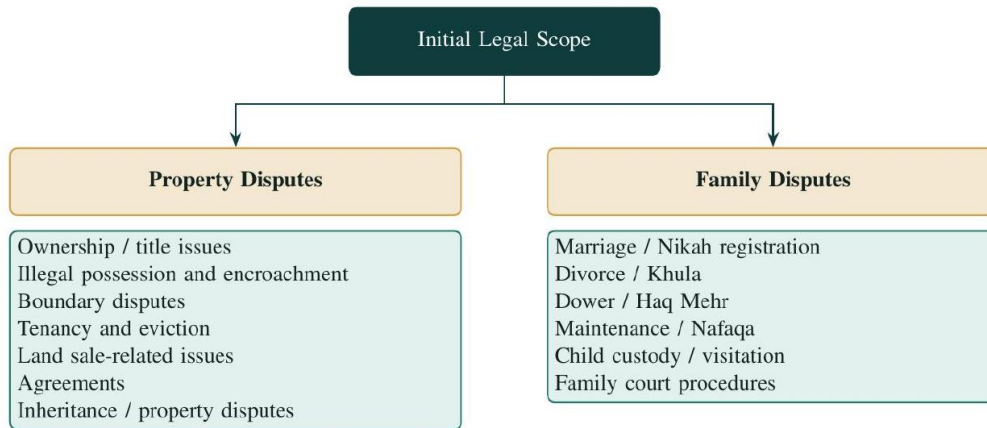


Figure 2: Initial legal scope — Property Disputes and Family Disputes

Table 2: Initial legal scope and topics covered

Area	Topics covered
Property Disputes	Ownership / title issues; illegal possession and encroachment; boundary disputes; tenancy and eviction; land sale-related issues; agreements; inheritance / property disputes.
Family Disputes	Marriage / Nikah registration; divorce / Khula; dower / Haq Mehr; maintenance / Nafaqa; child custody / visitation; family court procedures.

1.1.8 Expected Benefits

- Citizens receive understandable legal information with traceable sources, know what evidence to prepare, and can find relevant lawyers more easily.
- Lawyers receive cases that are better prepared and better documented.
- Administrators gain a single place to manage users, lawyers, cases and legal content.

The platform supports legal understanding and does not replace qualified lawyers. Matters that need professional judgment should always be reviewed by a lawyer.

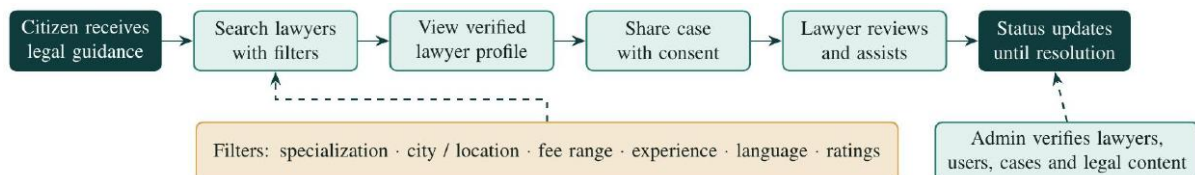


Figure 3: Lawyer discovery and case progress

Figure 3 shows the lawyer discovery process: after receiving guidance, the citizen searches lawyers using filters (specialization, city, fee range, experience, language and ratings), views

a verified profile, shares the case with consent, and receives status updates until resolution. The administrator verifies lawyers, users, cases and legal content throughout.

1.2 Existing Examples / Solutions

Two existing Pakistani platforms are relevant to this project: MyCounsel and QanoonAI. The descriptions below are based on their public descriptions (the Stanford CodeX TechIndex entries for each product, last reviewed July 2026). The team has not tested these products itself; features that the public descriptions do not mention are therefore marked as “not stated” rather than assumed to be absent.

Competitor 1: MyCounsel

Table 3: Profile of MyCounsel

Aspect	Description
Background	A Pakistani legal platform combining a lawyer marketplace with AI legal research tools. It serves both consumers and law firms.
Purpose	To make it easier to find lawyers and to support professional legal work online.
Main functionality	Clients can search verified lawyers, book paid consultations, and manage engagements, messages and documents online. Firms and in-house teams can use shared workflows, centralized billing and AI research over Pakistani case law and law journals.
How it works	The user searches the lawyer marketplace, books a consultation and continues the engagement through the platform, while professional users use the AI research tools.
Strengths	Verified lawyer search, online booking and engagement management, and AI research over Pakistani legal material.
Limitations	The public description does not mention fact-clarifying follow-up questions for a citizen’s case, evidence checklists, OCR, or a retrieval-grounded explanation of the user’s situation with citations.

Competitor 2: QanoonAI

Table 4: Profile of QanoonAI

Aspect	Description
Background	An AI legal intelligence platform focused on Pakistani law, aimed at judges, lawyers and citizens. Citizens have free access and professional users have paid plans.
Purpose	To give faster access to Pakistani judgments and legal research support.
Main functionality	A database of Pakistani judgments with tools for neutral case briefs, petition drafting, citation verification, court-fee calculation and Urdu-language legal guidance.
How it works	Users query the judgment database and use the drafting, verification and calculation tools.
Strengths	Pakistani legal focus, judgment database, citation verification and Urdu-language support.

Aspect	Description
Limitations	The public description is centered on judgments and professional tools. It does not mention a lawyer directory, case tracking, evidence checklists or OCR.

Comparison

Table 5: Comparison of existing solutions and the proposed system. “Not stated” means the public description does not mention the feature.

Feature	MyCounsel	QanoonAI	Proposed System (planned)
Legal information	AI research on case law and journals	Judgments, case briefs	Property and family law information
Pakistani legal focus	Yes	Yes	Yes
Case-based analysis	Not stated	Case briefs	Analysis of the user’s case facts
RAG / grounded responses	Not stated	Not stated	Yes
Source citations	Not stated	Citation verification	Yes
Follow-up questions	Not stated	Not stated	Yes
Document processing	Online document sharing	Petition drafting	PDF text and clause extraction
OCR	Not stated	Not stated	Yes
Evidence checklist	Not stated	Not stated	Yes
Lawyer directory	Yes (verified lawyers)	Not stated	Yes (verified lawyers)
Lawyer filtering	Search	Not stated	Specialization, city, fee, experience, language, rating
Case tracking	Engagement management	Not stated	Yes

Main gap. The reviewed platforms focus either on lawyer discovery and professional research, or on judgments and legal drafting. The proposed system addresses the gap for an ordinary citizen: a single case-centered workflow that clarifies facts, retrieves verified sources, explains them in plain language with citations, checks the needed evidence, connects the user to a suitable lawyer, and tracks the case, within the focused areas of property and family disputes.

1.3 Business Scope

This section describes who the platform is intended for and how it could be developed beyond the FYP. The commercial points are possibilities for the future; none of them is confirmed revenue.

Table 6: Business scope of the platform

Item	Description
------	-------------

Item	Description
Target market	Pakistani citizens, especially people facing property or family disputes, people who want understandable legal information, and people looking for relevant lawyers.
Additional users	Lawyers, legal professionals and relevant legal organizations, who can use professional profiles and receive better-prepared cases.
Business potential	Possible future opportunities: premium services for citizens, lawyer discovery and matching, professional lawyer profiles, subscription-based services, and organization-level access for legal organizations.
Commercial deployment	After the FYP, the platform could be deployed as a web service with the same portals for citizens, lawyers and administrators. Commercial use would first require further legal review of the knowledge base, stronger security and operational support.
Scalability	The backend, the application database and the vector database are separate components, so each can be scaled or replaced independently as users and documents grow.
Future expansion	The same architecture can later support more legal domains by adding verified sources to the knowledge base. No additional domain is part of this FYP.

1.4 Useful Tools and Technologies

Table 7 lists the technologies selected for the project, what each is used for, and why it was chosen.

Table 7: Technology stack

Technology	Purpose	Reason for selection
Python	Backend logic, AI/RAG pipeline, document processing	Strong libraries for NLP, embeddings and PDF processing
JavaScript / TypeScript	Frontend development	Used with React / Next.js to build the portals
React / Next.js	Responsive citizen, lawyer and admin portals	Component-based UI and good support for responsive web applications
FastAPI	APIs, authentication, case workflows, RAG integration	Python framework that works directly with the AI code; fast REST APIs
MongoDB	Application database: users, lawyers, profiles, case information	Flexible document model suits profiles, cases and varying case details
Qdrant	Vector database: legal-document embeddings, semantic search, similarity retrieval	We chose Qdrant for fast and accurate semantic search in our RAG system.
Sentence Transformers	Embeddings of legal text and user queries	Open-source and widely used for semantic retrieval
Open-source LLM	Grounded response generation from retrieved sources	Can be run and tested without relying on a paid service
PyMuPDF	Text extraction from PDF documents	Fast and practical for PDF text and layout
OCR	Text from scanned documents and images	Needed for scanned legal documents and user-uploaded evidence

Application Database and Vector Database

The two databases have different roles. MongoDB stores application data such as users, lawyers, profiles and cases. The vector database stores only the legal-document embeddings (numerical representations of the text) and is used for semantic retrieval in the RAG pipeline. Keeping them separate lets each be optimized, scaled or replaced independently.

System Architecture

Figure 4 shows the three-layer architecture. The presentation layer (React / Next.js) contains the citizen, lawyer and admin portals. The application layer (FastAPI) contains authentication and user management, case workflow and tracking, document processing (PyMuPDF and OCR), query and follow-up handling, and the RAG service. The data layer contains MongoDB, the uploaded-document store and the vector database. The layers communicate over HTTPS using REST APIs.

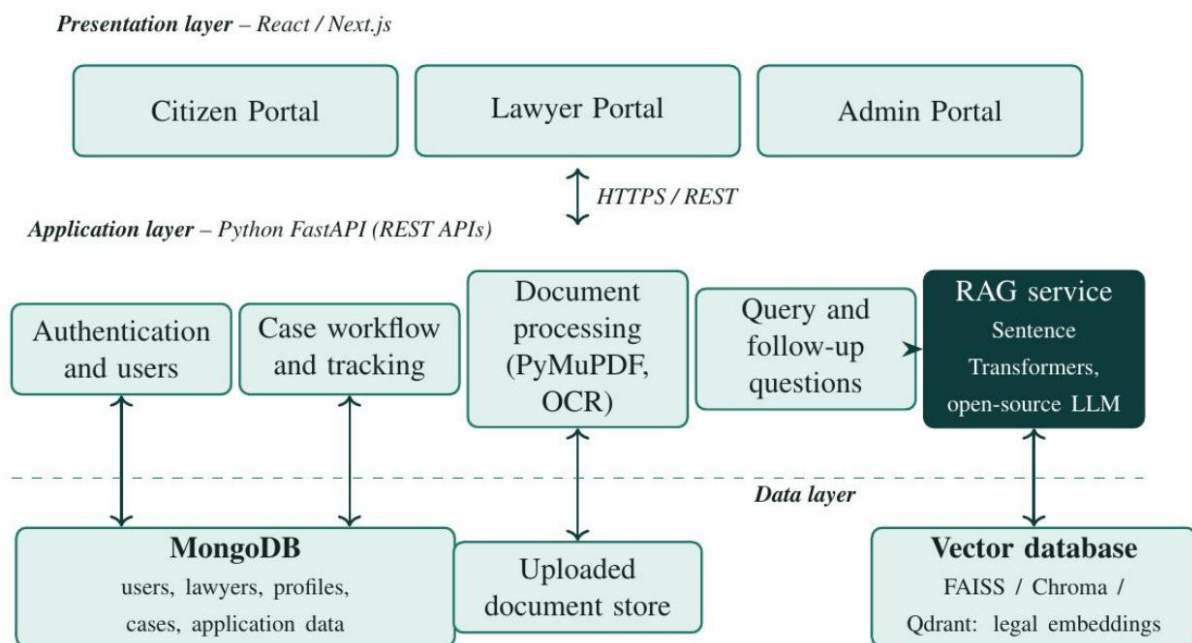


Figure 4: System architecture

RAG Pipeline

Figure 5 shows how the RAG pipeline works in two phases.

- **Offline phase (building the knowledge base).** Verified Pakistani legal sources are collected, cleaned and verified; text is extracted (PyMuPDF, OCR); the text is split into section-wise chunks with metadata; embeddings are created with Sentence Transformers; and they are stored in the vector database.
- **Online phase (answering a case).** The user's case description and answers are received; follow-up questions fill any missing facts; the most relevant sources are retrieved semantically; a grounded prompt is sent to the open-source LLM; the answer is checked against its sources and citations; and a plain-language answer with citations is returned.

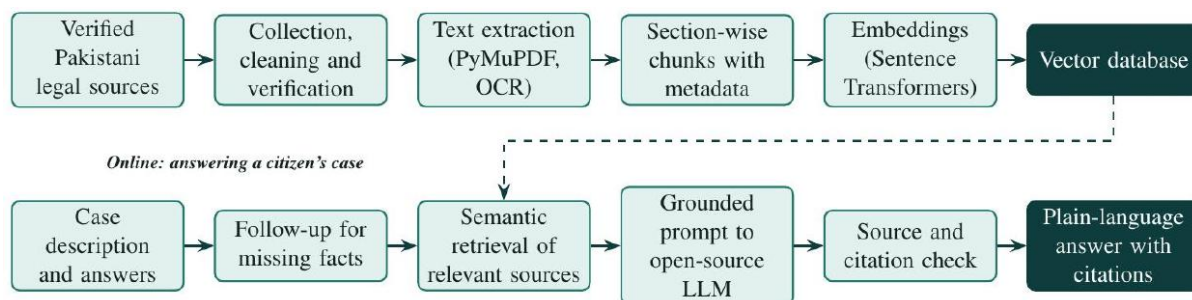


Figure 5: RAG pipeline — offline knowledge base preparation and online answering

Development Environment, Operating Systems and Network Protocol

- **Development environment.** Code editors and Git-based version control for team collaboration, together with Python virtual environments and Node.js for the backend and frontend.
- **Operating systems.** Development is possible on Windows, Linux or macOS because Python, Node.js and MongoDB are cross-platform. Deployment is planned on a Linux-based server.
- **Network protocol.** The frontend and backend communicate over HTTP/HTTPS using REST APIs. HTTPS protects data in transit.

1.5 Project Work Breakdown

The project is divided into 20 work packages (Table 8). Each package has a clear description so that progress can be tracked and responsibilities assigned.

Table 8: Work breakdown structure

No.	Work package	Description
1	Project planning	Define scope, roles, schedule and tools.
2	Requirements analysis	Collect user, lawyer and admin requirements and prepare the feature list.
3	Legal research	Study the property and family dispute areas to understand the legal issues involved.
4	Legal source identification	Select the verified Pakistani legal sources to be used.
5	Legal corpus collection	Collect the selected sources into one corpus.
6	Data cleaning and verification	Clean the text and check the reliability of each source.
7	Document processing	Extract PDF text with PyMuPDF and prepare OCR for scanned files.
8	Vector database / embedding preparation	Split text into chunks, create embeddings and store them in the vector database.
9	RAG baseline	Build the first working retrieval and grounded generation pipeline with citations.
10	Backend development	Develop FastAPI services, authentication and MongoDB integration.
11	Frontend development	Develop the citizen, lawyer and admin portals.

No.	Work package	Description
12	Case workflow	Implement case creation, follow-up questions and status tracking.
13	Lawyer directory	Implement lawyer profiles, verification, search and filters.
14	Document / OCR integration	Integrate upload, text extraction and the evidence checklist into the application.
15	System integration	Connect the frontend, backend, databases and RAG service.
16	Testing	Perform functional and system testing and report bugs.
17	Security testing	Check access control, authentication and data protection.
18	Evaluation	Compare the standard LLM with the RAG system and measure usability with SUS.
19	Deployment	Set up the final system on a server.
20	Documentation and defense preparation	Prepare reports, presentation and defense.

Team Assignment

The team works collaboratively while each member keeps a primary responsibility (Table 9).

Table 9: Team assignment

Member	Primary responsibility	Collaborative work
Muhammad Nabeel	Development: full-stack, backend/frontend integration, RAG and system integration	Research, data, integration, testing, documentation
Ashir Qureshi	Development: full-stack, frontend/backend development, feature implementation and integration	Research, data, integration, testing, documentation
Muhammad Sharjeel	QA/Testing: test planning, functional and system testing, bug identification, quality assurance	Research, data, testing, documentation, integration support

All three members take part in research, requirements gathering, legal-domain research, legal source identification, legal data collection, data cleaning and verification, competitor analysis, feature planning, system and UI/UX discussions, documentation, integration, testing discussions, problem solving, and preparation of the final presentation and defense.

1.6 Project Timeline

The project is planned over seven months (M1 to M7). The phases are:

- **M1–M2:** legal research, source identification, corpus collection, and cleaning and verification.
- **M3:** document processing, embeddings and the RAG baseline.
- **M4–M5:** backend, frontend, case workflow, lawyer directory and system integration.
- **M6:** security testing and evaluation.
- **M7:** deployment and defense.

Exact dates are not fixed yet. Planning, requirements and documentation run alongside the technical work, and testing begins as soon as the first features are integrated.

Table 10: Seven-month project timeline (shaded cells show active work)

No.	Task	M1	M2	M3	M4	M5	M6	M7
1	Project planning							
2	Requirements analysis							
3	Legal research							
4	Legal source identification							
5	Legal corpus collection							
6	Data cleaning and verification							
7	Document processing							
8	Vector database / embedding preparation							
9	RAG baseline							
10	Backend development							
11	Frontend development							
12	Case workflow							
13	Lawyer directory							
14	Document / OCR integration							
15	System integration							
16	Testing							
17	Security testing							
18	Evaluation							
19	Deployment							
20	Documentation and defense preparation							